

# PS15\_SolarFlare — Summary

PS15\_SolarFlare â€” SoLEXS Nowcasting (Aditya-L1)

## Summary

- Repository for SoLEXS-only nowcasting pipeline developed for a Hack2Skill / ISRO submission (BAH 2026 PS#15).

## Data sources

- SoLEXS combined lightcurves: local (pre-provided) files under `data/` and combined into `output/solexs\_combined.csv`.  
- GOES X-ray event list (for labeling): NOAA NCEI/NGDC GOES event lists (public domain). Example source page: <https://www.ngdc.noaa.gov/stp/satellite/goes/>. We used `data/goes\_xray\_2026.txt` as the event-list file and included attribution in `scripts/label\_flares.py` outputs.

## Notes on competition rules

- NOAA GOES data is public-domain U.S. government data and is generally permissible to use in hackathons; however, always confirm the specific hackathon's dataset/external-data rules.  
- Hack2Skill resources: <https://hack2skill.com> and the Hack2Skill YouTube channel: <https://www.youtube.com/@hack2skill>

## Repro steps

1. Fetch or place GOES event list: `data/goes\_xray\_2026.txt` (optional: run `py scripts/fetch\_goes.py 2026` to attempt download).
2. Combine SoLEXS FITS (already done): `output/solexs\_combined.csv`.
3. Label SoLEXS with GOES events: `py scripts/label\_flares.py` (produces `output/labeled\_solexs.csv` and `output/matched\_flares.csv`).
4. Generate plots: `py scripts/plot\_solexs.py` (produces `output/solexs\_plot\_YYYYMMDD.png` and `output/solexs\_plot\_zoom.png`).

## Attribution

- GOES X-ray event lists: NOAA NCEI/NGDC â€” <https://www.ngdc.noaa.gov/stp/satellite/goes/>

If you want, I can also generate a short 1â€”2 page PDF summary for submission and include these citations in the PDF. Please confirm and I'll produce it.



